

Module-1: Nanomaterials

Nanomaterial

It is defined as the material with any external dimension in the nano-scale which ranges from 1 nm to 100 nm. This includes both nano-object, discrete pieces of material and nano-structural materials.

Sources of Nanomaterial

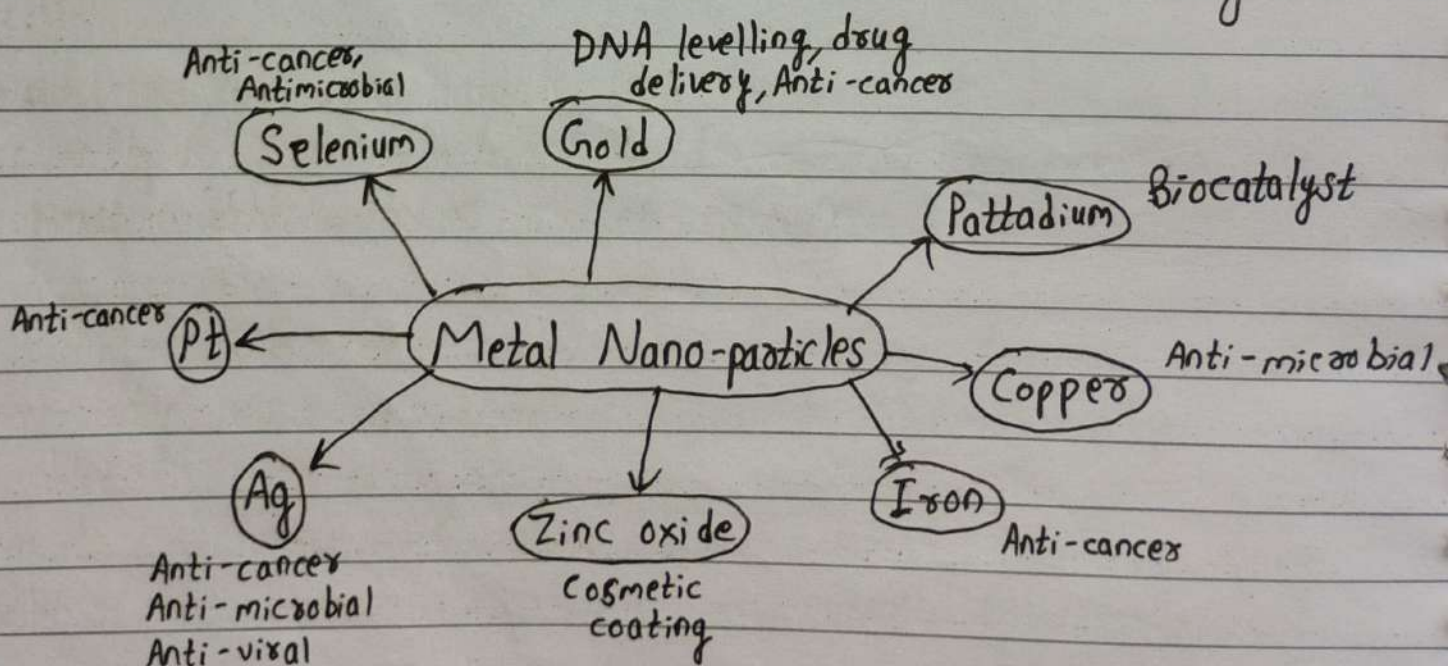
1. Engineered

1. → Carbon Black and Titanium dioxide (TiO_2)

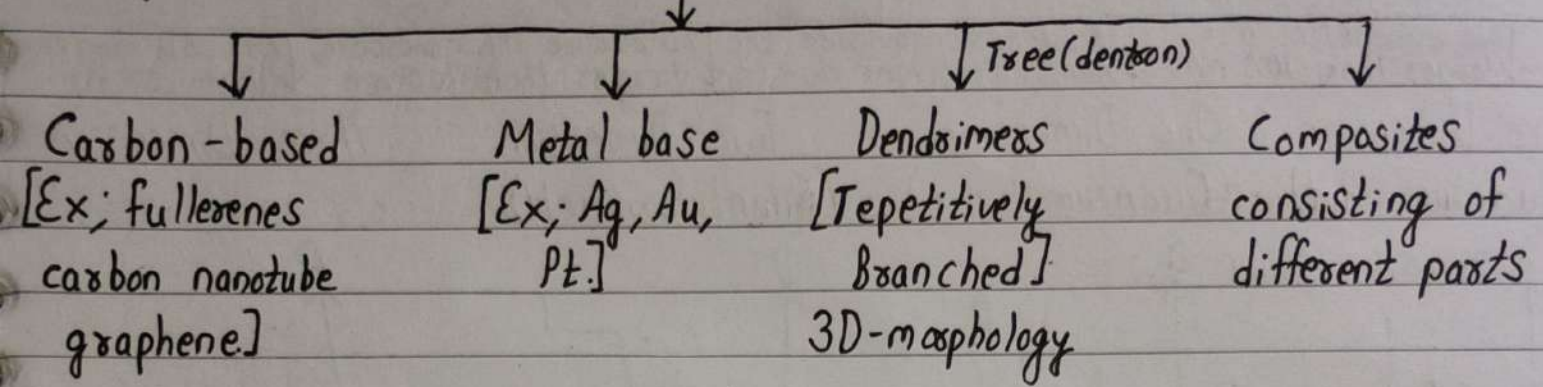
2. → Incidental → produce as by-products of mechanical and industrial process. For ex: vehicle-engine exhaust, welding fumes, combustion of solid fuels.

3. Natural

→ Natural sources of nano-material are; forest fire product, volcanic dust, radioactive decay of radon gas.

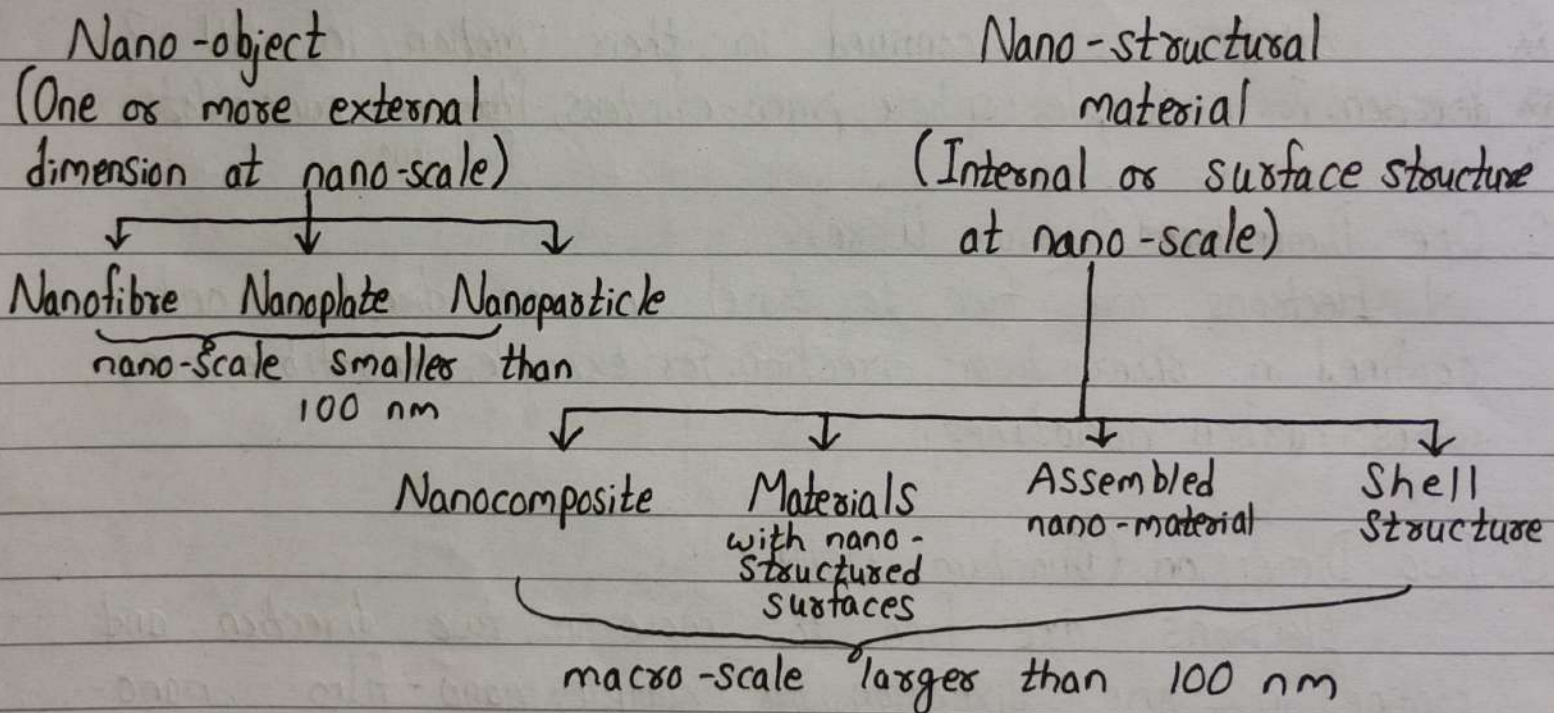


Types of Nano-materials (Engineered Nanomaterials)



Types of nano-material

[External or internal dimension at nano-scale]

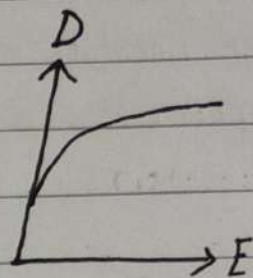
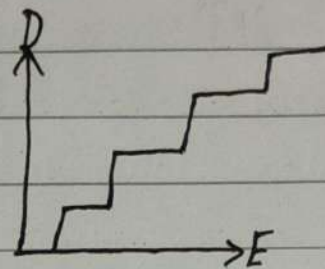
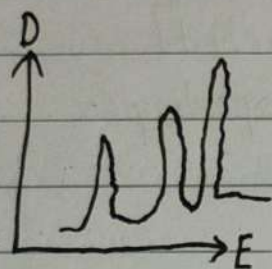
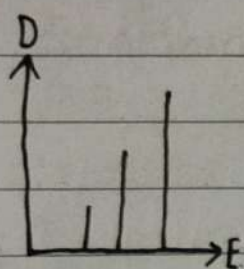


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Classification of Nano-materials on the basis of Dimension

No dimensions are larger than 100 nm 1D dimension outside the nanoscale, larger than 100 nm 2D outside the nanoscale, larger than 100 nm All 3D, larger than 100 nm

Zero Dimension (Quantum dot) One Dimension (Quantum wire) Two Dimension (Quantum well) Three Dimension



1. Zero Dimension (Quantum Dot):

Electrons are confined in their motion in all 3 direction. For example; sphere, nano-clusters, ~~fluorene~~ nano-dots, fullerenes

2. One Dimension (Quantum Wire):

Electrons are free to travel in one direction and confined in other two direction. For example; nano-fibre, nano-wires, carbon-nanotube.

3. Two Dimension (Quantum Well):

Electrons are free to move in two direction and confined in one direction. For example; nano-film, nano-plates, graphene

4. Three Dimension:

Electrons are free to move in all three directions and there are no-confinement and limitations. For example; powder, multi-layer fibres and poly-crystalline materials.

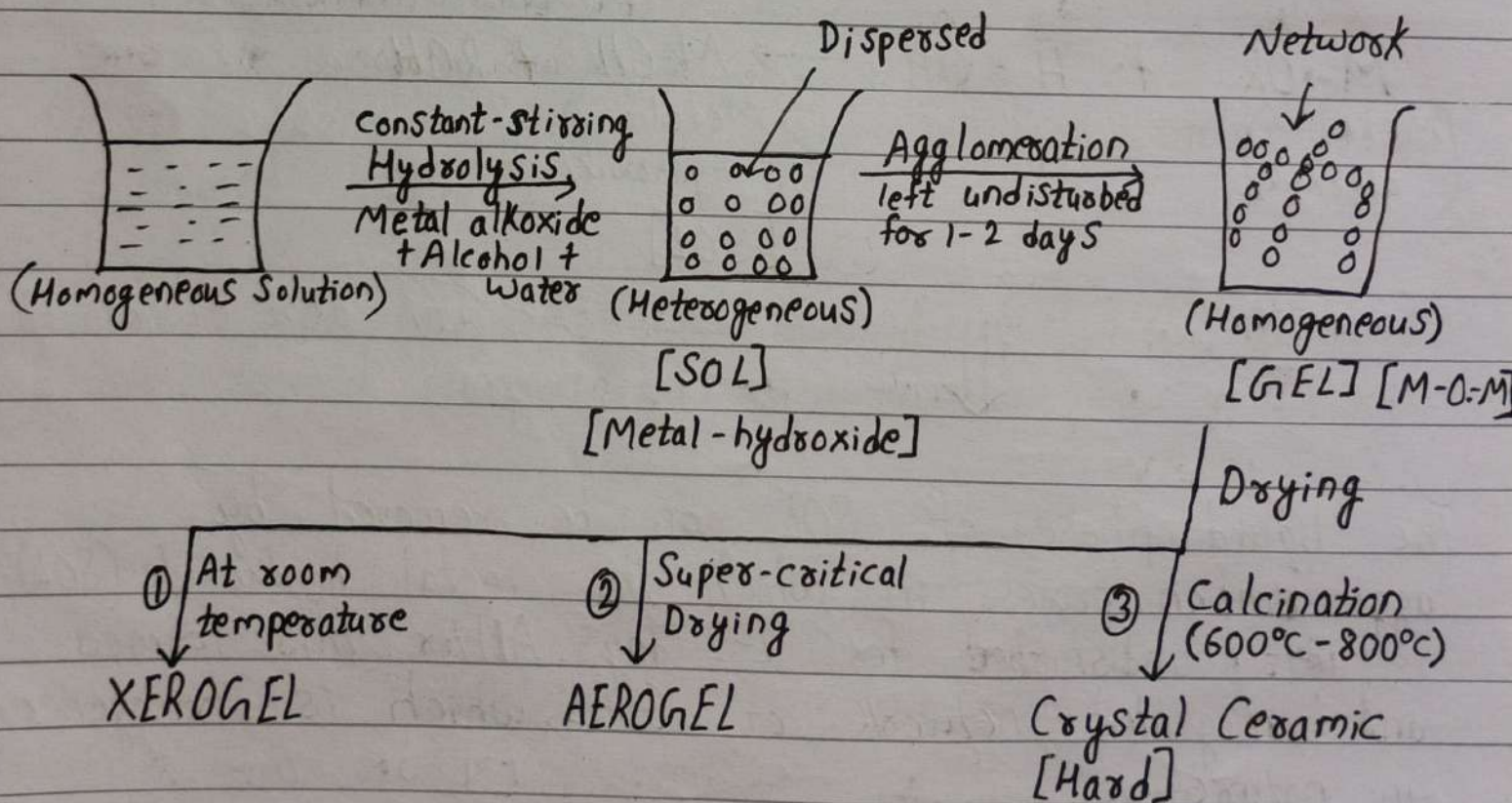
* The nano-structural of 0-D, 1-D & 2-D are closely related with each other and formed interfaces. For example; diamond & graphite.

Applications of Nano-materials

- Automotives
- Engineering
- Medicine
- Cosmetics
- Textiles
- Sports
- Chemicals
- Electronics

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SOL - GEL Method

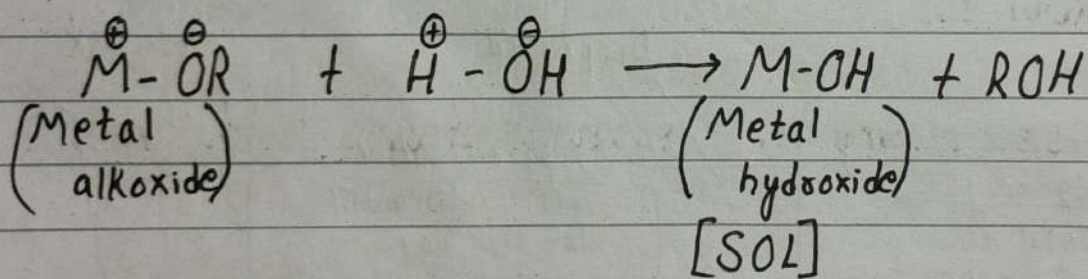


→ SOL-GEL process involves the transition of solution system from liquid SOL to solid SOL.

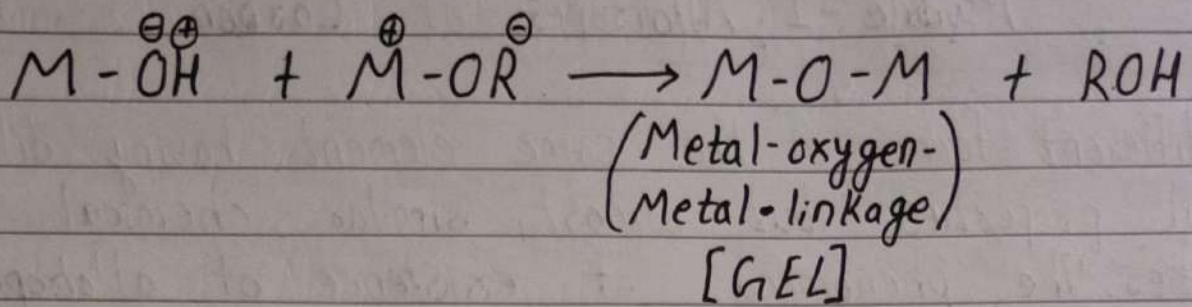
→ It is a wet-chemical technique widely used for the fabrication of nano-structured ceramic materials and thin films.

→ The SOL-GEL method is a bottom-up chemical synthesis technique.

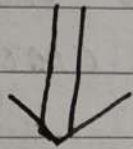
The SOL-GEL prepared with the help of metal-alkoxide or metal chloride, hydrolysed by adding alcohol and by constant stirring to form metal hydroxide which is heterogeneous in nature. This metal hydroxide is also called colloidal suspension. It is also known as SOL.



The liquid phase of SOL can be removed by agglomeration process in which the metal hydroxide (SOL) is left undisturbed for 1-2 days. After this, formed and integrated network of GEL, which is homogeneous in nature.



Formation of metal oxide with oxo ($M-O-M$) and hydroxo ($M-OH \cdots M$) bridges, formed polymer in the solution.



Drying Process: GEL can be drying at different phases that is. (i) at evaporate at room temperature to formed XEROGEL.

(ii) GEL can be drying at super-critical condition to formed AEROGEL.

(iii) GEL can be drying at heating $600-800^{\circ}C$ (Calcinate) to formed Crystal Ceramic.